

The blocking criterion is monotonically worse in the block size: blocking gains nothing without a change of coupling

Reading Yarotsky's proof (Commun. Math. Phys. 261 (2006) 799, §2, passage level) settles the input left open by the blocking note: the proof is a polymer expansion, its Lemma 1 bounds the non-classical evolution of an excited region I by $(2\alpha e^{t_0\beta/\alpha})^{|I|}$ and its Lemma 3 gives configuration weights exponentially damped in the size of the excited region, so the connected structure that the blocking step needed is present, and the criterion applies to a lattice of blocks with the block Hamiltonians as the classical part. The conclusion is negative and sharp. With blocks of M^3 sites the criterion reads

$$\beta_{\text{block}} = \frac{\text{straddling norm per block}}{\text{block gap}} = \frac{24M^2N}{g^2 \delta(g; M)} \leq \beta_*,$$

and $\delta(g; M)$ degrades or stays flat as M grows: at strong coupling the block gap is the single flux-loop energy $2C_2g^2$, independent of M , so $\beta_{\text{block}} \propto M^2$; in the small-volume regime $\delta \simeq \delta_1 g^{2/3}/M$, so $\beta_{\text{block}} \propto M^3$. Against the direct criterion $\beta_{\text{direct}} = 32N/(C_2g^4)$ of the T2 note the ratio is $\beta_{\text{block}}/\beta_{\text{direct}} \simeq \frac{3}{8}M^2$, already above one at $M = 2$. **Real-space blocking with a criterion of this form is strictly worse than no blocking at all**, at every coupling, and iterating it cannot reach weaker coupling. This corrects Sections 4 and 5 of the blocking note, which assumed that the connected estimate would buy a fixed threshold and an induction: the estimate is available, and it buys nothing, because the boundary grows like M^2 while the gap it is measured against does not grow at all. Any gain must come from recognizing the blocked Hamiltonian as a theory of the same form with a **larger effective coupling**, which is the renormalization step itself. Constants explicit; nothing promoted.

1. The connected structure is present in the proof

Yarotsky's Theorem 1 is proved by writing

$$e^{-t_0 H_\Lambda} = \sum_{I \subset \Lambda} T_{\Lambda, I}, \quad T_{\Lambda, I} = \sum_{J \subset I} (-1)^{|I|-|J|} e^{-t_0 (H_{\Lambda, 0} + \sum_{x \in J} \phi_x)},$$

so that $T_{\Lambda, I}$ collects the contributions in which every site of I is touched by the perturbation. Two estimates carry the argument (§2, passage level).

- **Lemma 1.** $\|T_I\| \leq (2\alpha e^{t_0\beta/\alpha})^{|I|}$, proved by analytic continuation of the couplings $\phi_x \rightarrow z_x \phi_x$, the Hille–Yosida bound $\|e^{-t_0 H_J(z_J)}\| \leq e^{t_0 |J| \beta/\alpha}$ on the numerical range, and a multidimensional Schwarz lemma using that $T_I(z_I)$ vanishes whenever any $z_x = 0$.
- **Lemma 3.** The weight of a space-time configuration $C = \{(I_k, J_k)\}$ obeys $|w(C)| \leq \prod_k (2\alpha e^{t_0\beta/\alpha})^{|I_k|} e^{-t_0 (|J_k| - |\Lambda_0|^3 |I_k|)}$, the classically excited sites contributing e^{-t_0} each through the local gap.

Both are exponential in the size of the excited region, which is exactly the polymer structure that makes the expansion sum over **connected** clusters. So the hypothetical estimate of the blocking note §3, replacing $(\sum_p \|w_p\|)^2$ by $\sum_p \|w_p\|^2$, is precisely what the expansion produces, and the way to use it is to apply the theorem itself to the blocked system in place of a re-derived Schur bound.

2. The criterion on a lattice of blocks

Identify the blocks of M^3 sites with the sites of a coarse cubic lattice, as in the blocking note §1. Take as classical part the normalized block Hamiltonians

$$h_\alpha = \frac{H_\alpha - E_0^\alpha}{\Delta_M}, \quad \Delta_M = \frac{\hbar c}{a} \delta(g; M),$$

each with non-degenerate ground state Ω_α (T1) and unit gap, and as perturbation the straddling plaquettes,

$$\phi_\alpha = \frac{1}{\Delta_M} \sum_{p \text{ straddling at } \alpha} w_p, \quad \|\phi_\alpha\| \leq \frac{6M^2 \cdot 4N\hbar c/(ag^2)}{\Delta_M} = \frac{24M^2N}{g^2 \delta(g; M)} =: \beta_{\text{block}}.$$

The perturbation is bounded, so $\alpha = 0$ in Yarotsky's condition (2) and the criterion is $\beta_{\text{block}} \leq \beta_*(3, \{0, 1\}^3)$, with the same constants as in the T2 note. The classical part is a single-block operator, hence a function of its own spectral partition of unity, and the partition contains the projection onto Ω_α ; the interaction range is one coarse lattice spacing.

3. The criterion degrades with M

Two regimes fix the behaviour of $\delta(g; M)$.

Strong coupling. The lowest gauge-invariant excitation of a block is a single plaquette flux loop, of energy $2C_2(R_{\min})g^2\hbar c/a$ independent of the block size, as in the obligations map §3. So $\delta(g; M) \rightarrow 2C_2g^2$ and

$$\begin{aligned} \beta_{\text{block}} &\simeq \frac{24M^2N}{2C_2g^4} = \frac{12M^2N}{C_2g^4}, \\ \frac{\beta_{\text{block}}}{\beta_{\text{direct}}} &\simeq \frac{12M^2N/(C_2g^4)}{32N/(C_2g^4)} = \frac{3M^2}{8}, \end{aligned}$$

using $\beta_{\text{direct}} = 32N/(C_2g^4)$ from the link-level check of the T2 note. At $M = 2$ the ratio is $3/2$, and it grows quadratically thereafter.

Small volume. For g small and a block of side Ma the gap is the zero-mode gap of G07 Proposition 7, $\Delta_M = \delta_1 g^{2/3} \hbar c/(Ma)$, that is $\delta(g; M) = \delta_1 g^{2/3}/M$. Then

$$\beta_{\text{block}} \simeq \frac{24M^3N}{\delta_1 g^{8/3}},$$

growing like M^3 and diverging as $g \rightarrow 0$.

In both regimes the numerator grows like the block surface, M^2 , while the denominator stays flat or shrinks. The criterion therefore has its best value at $M = 1$, which is the direct application already used for T2.

Proposition. For every $M \geq 2$ and every $g > 0$, $\beta_{\text{block}}(M) \geq \beta_{\text{block}}(1)$ whenever $\delta(g; M) \leq M^2\delta(g; 1)$, which holds in both regimes above and follows in general from the upper bounds of the Polyakov note, where the gap of a box of side L is at most of order $\hbar c/L$ times a coupling factor. Blocking therefore never improves the criterion.

4. What this rules out, and what it leaves

The negative result is specific: **a criterion that compares the total inter-block coupling with the block gap cannot be improved by blocking**, because coupling scales with area and the gap does not scale with volume. It rules out the naive induction of the blocking note §5, in which one hoped that a fixed threshold, once reached, would propagate to all larger scales.

What survives is the observation that made the induction attractive: the effective coupling of an asymptotically free theory grows toward the infrared. To use it, the blocked Hamiltonian must be **recognized as a theory of the same type with a new coupling**, so that the criterion is re-applied with $g_{n+1} > g_n$ and with the block again playing the role of a single site of unit spacing. That identification is the renormalization-group step; it requires a change of variables at each scale, not a projection. Yarotsky's own Theorem 3 does precisely this in its setting: he treats the AKLT model by showing that *on a large length scale it is a relatively bounded perturbation of a classical model*, which works because the frustration-free structure makes the block ground state exactly known. The Kogut–Susskind Hamiltonian at intermediate coupling has no such structure, and supplying a substitute is the open problem.

5. Correction to the previous note

The blocking note §§4–5 concluded that a connected estimate would give a one-step inequality valid above a fixed threshold $g^4 \gtrsim 8\sqrt{6}CMN/(\gamma C_2)$ and that the induction would then close. Sections 1–3 above show that the connected estimate is available and that the resulting threshold is worse than the direct one by the factor $3M^2/8$, so the induction does not close by this route. The finite-step counting of §5 of that note, $n \simeq (2b_0 \log 2)^{-1}(g_{\text{UV}}^{-2} - g_{\text{thr}}^{-2})$, retains its meaning as a count of renormalization-group steps, with the proviso that each step must redefine the coupling rather than merely project.

6. Consequence for STATE

The blocking route in its projection form is closed, with the reason stated quantitatively: boundary grows like M^2 , gap does not grow. The route survives only in its renormalization form, where each step redefines the theory, which is the constructive problem. The nearest remaining tractable questions are on the other side of the ledger: the stability theorem without frustration-freeness named in the Lieb–Robinson note §5, and the fixed-lattice small-volume theorem (1a) named in the Schur note §5.